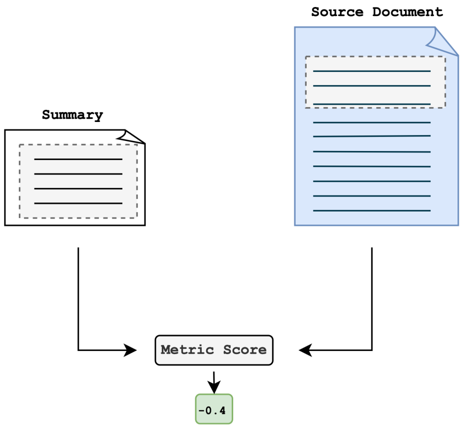

Figure 2: Calculation of a traditional automatic metric for assessing factual consistency.

a long document, it will be ignored, which is a problem when assessing factual consistency of long document summarisation.

- The second significant difference is that LongDocFACTScore calculates a metric score on short sections of text at one time, comparing one sentence in the predicted summary to a short section of the source document, rather than long, truncated sections. This allows for better evaluation of fine-grained statements.
- Lastly, LongDocFACTScore uses sentence embeddings to identify the most similar parts of the source document. This improves the efficiency of the framework, as it avoids the metric needing to be applied for each pair-wise set of sentences in the source document and summary.

4. Experimental Data Sets

We evaluate the automatic metrics on their ability to assess factual consistency on two long document data sets and several short document data sets. We collected our own long document data set, consisting of documents from the biomedical and scientific domains annotated by six expert human annotators with fine-grained factual consistency labels. We refer to this data set as the LongSciVerify data set and provide further details of its curation in Section 4.1. The data set is made available alongside our code. We further evaluate our methods on the LongEval PubMed data set (Krishna et al., 2023), another long document data set with factual consistency annotations. Finally, we conduct an evaluation on a range of short document data sets with human annotations of factuality, which have been used to evaluate automatic metrics in prior works (Yuan et al., 2021).

	Doc. tokens	Doc. sentences	Sum. tokens	Sum. sentences
PM	3209	124	208	9
AX	6515	249	279	11

Table 1: Average number of tokens and sentences in the evaluated data sets. PM denotes the PubMed data set and AX denotes the ArXiv data set.

4.1. The LongSciVerify Data Set

To support the evaluation of factuality metrics for long documents, we create a new data set called LongSciVerify, with multiple summaries generated from long documents, and fine-grained human annotation scores of their factual correctness. This data set consists of 270 annotated summaries generated from the long document, English-language PubMed and ArXiv data sets (Cohan et al., 2018). A description of the PubMed and ArXiv data sets can be found in Table 1.

From each of the PubMed and ArXiv data sets, fifteen articles were randomly sampled. Summaries were generated for these data sets using three different abstractive methods, which were all able to consider the entire long document in the generation of their summaries. These methods were selected to enable an effective evaluation of the performance of the automatic metrics in long document settings. Details of the abstractive methods used to generate the summaries are provided in Appendix A. As the PubMed and ArXiv data sets included in this data set are highly domain specific, we recruited six expert annotators, three per data set, to review the automatically generated summaries. At the time of evaluation, all of the expert annotators reviewing the PubMed data set were, or were in the final years of study to be, qualified clinicians. The expert annotators for the ArXiv data set had all achieved a minimum of an undergraduate degree in a physical science. The annotators who participated in our study were colleagues of the authors and therefore volunteered to participate in the study without payment. It was made clear to the annotators that this human evaluation was for scientific research on abstractive summarisation with the intention for use in a scientific publication.

The definition of factual consistency we provided to annotators was taken from Fabbri et al., (2021): *"Factual consistency: The factual alignment between the summary and the summarised source. A factually consistent summary contains only statements that are entailed by the source document."*.

We opted to capture a fine-grained binary classification metric (entailed vs not entailed), due to this having been shown to be effective and achieve higher inter-annotator agreement scores (IAA) in prior work (Krishna et al., 2023; Min et al., 2023).

	LSV ArXiv	LSV PubMed	LE PubMed
Fine-grained	0.54	0.76	-
Summary-level	0.70	0.82	0.61

Table 2: IAA of the human-annotated data for the LongSciVerify (LSV) and LongEval (LE) data sets, calculated using Krippendorf's alpha metric, for fine-grained and summary-level annotations of factual consistency.

Annotators were asked to mark a sentence as 'not entailed' if there were any factual inconsistencies. For each generated summary included in the study, we sampled three summary sentences and selected the most similar two text snippets (1-3 sentences) from the source document, calculated using sentence embeddings and cosine similarity. The human annotators were then given the three sentences sampled from the generated summary, and the corresponding two text snippets for each, from the source document and were asked to decide whether, given the text snippets, if each sentence was entailed or not. We provide an example screenshot of the factuality scoring for the three sampled sentences from a PubMed article summary in Figure 3.

For each of the PubMed and ArXiv samples, each of the three human annotators evaluated the same three summaries (generated by the three different methods) from the same 15 randomly sampled documents, thus resulting in 270 annotated summaries, with three fine-grained annotations per summary. During the human evaluation study, the annotators were unaware of which method was used to create each summary.

Table 2 shows the IAA of the fine-grained human annotated data, for each data set, calculated using the Krippendorff's alpha metric[4] (Krippendorff, 2004). In Table 2, the IAA is calculated both between fine-grained, sentence-level entailment annotations, and for the summary-level annotations (i.e., the average of the fine-grained annotations per summary). For our LongSciVerify PubMed data set, the IAA of the fine-grained factual consistency annotations is relatively high. However, the IAA of the LongSciVerify ArXiv data set is a little lower. We hypothesise this could be due to the noise in the ArXiv data set (Koh et al., 2022) and the highly domain-specific nature of the data set.

4.2. The LongEval Data Set

We additionally evaluated LongDocFACTScore and the other automatic metrics included in our study on the publicly available long document PubMed LongEval data set (Krishna et al., 2023). This data set consists of summaries generated from two abstractive models: LongT5-large (Guo et al., 2022) and BigBird-PEGASUS (Zaheer et al., 2020). Three expert annotators were hired to give annotations of factuality on 40 summaries (two summaries generated by different methods for 20 documents), giving 120 annotated summaries. The IAA of the summary-level human annotations of factual consistency is given in Table 2. As for the LongSciVerify data set that we create, the LongEval data set was created by averaging fine-grained annotations to give a summary-level factuality score. However, in constrast to the LongSciVerify data set, annotators considered the entire source article, rather than just the most relevant sections of it, when making their assessments.

5. Experiments

5.1. Experimental Setting

As baselines, ROUGE[5] and BERTScore were implemented. We additionally implemented SOTA reference-free metrics: FactCC[6], QuestEval[7], and BARTScore[8] (using the 'bart-large' model[9]). In this work, we apply the LongDocFACTScore framework to extend the state-of-the-art (SOTA) metric BARTScore, also implemented with the 'bart-large' model. All experiments were run on a single NVIDIA v100 GPU and all metrics, apart from ROUGE, made use of the GPU compute. For the long document data set evaluation, all metrics were applied in a reference-free setting, i.e., comparing the predicted summary to the source document.

5.2. Long Document Data Set Results

To calculate the correlations between the human measure of factual consistency and automatic metrics, the fine-grained annotations were averaged to give a summary-level score. The summary-level human-annotated factuality scores were then averaged over the different annotators for each unique summary, thus giving a single summary-level human factuality score for each unique summary. These human-annotated, summary-level scores were then compared to summary-level scores predicted by each metric. Consequently, for each pair

[4] https://github.com/grrrr/krippendorff-alpha

[5] https://huggingface.co/spaces/evaluate-metric/rouge

[6] https://github.com/salesforce/factCC

[7] https://github.com/ThomasScialom/QuestEval

[8] https://github.com/neulab/BARTScore

[9] https://huggingface.co/facebook/bart-large

Randomly selected sentence from summary	Most similar sentences from source document	Your rating
for more subsequent investigations, the fimh gene was detected in upec strains isolated from hospitalized and out - patients with uti	0. in addition , single - nucleotide polymorphism (snp) analysis of fimh is a screening tool for epidemiological typing of upec (11 , 12) . therefore , the research on bacterial virulence factors can result in expansion and development of new methods for diagnosis and prevention of utis . for more subsequent investigations , the fimh gene was detected in upec strains isolated from hospitalized and out - patients with uti , referred to educational hospitals of shahrekord . 1. for more subsequent investigations , the fimh gene was detected in upec strains isolated from hospitalized and out - patients with uti , referred to educational hospitals of shahrekord . the present study was conducted for detection of the fimh virulence gene from upec isolated from both hospitalized patients and outpatients with uti , referred to educational hospitals of shahrekord , iran .	E
the fimh gene was detected in 98% of the e. coli isolates from uti.	0. for example , kaczmarek et al . (13) evaluated and detected the genes encoding virulence factors among e. coli strains with k1 antigen as well as the non - k1 e. coli strains . they found that the fimh gene existed in the whole tested e. coli k1 strains as well as in 97.0% of non - k1 strains . 1. (16) studied 18 upec isolates collected from females and found that the fimh gene was the most prevalent virulence factor and 100% of the isolates had that gene . in another study , 17) demonstrated that the fimh gene was the most frequent virulence gene and was detected in 98% of e. coli strains isolated from patient with utis .	E
we evaluated the virulence of the fimh gene in upec isolates from hospitalized patients and out - patients with upec.	0. in addition , single - nucleotide polymorphism (snp) analysis of fimh is a screening tool for epidemiological typing of upec (11 , 12) . therefore , the research on bacterial virulence factors can result in expansion and development of new methods for diagnosis and prevention of utis . for more subsequent investigations , the fimh gene was detected in upec strains isolated from hospitalized and out - patients with uti , referred to educational hospitals of shahrekord . 1. for more subsequent investigations , the fimh gene was detected in upec strains isolated from hospitalized and out - patients with uti , referred to educational hospitals of shahrekord . the present study was conducted for detection of the fimh virulence gene from upec isolated from both hospitalized patients and outpatients with uti , referred to educational hospitals of shahrekord , iran .	NE

Figure 3: Example of a PubMed summary, annotated for factual consistency by an expert human annotator to create the LongSciVerify data set. "E" indicates that a sentence is "Entailed" and "NE" indicates a sentence is "Not Entailed".

Metric	PubMed	ArXiv
ROUGE-1	0.09	0.02
ROUGE-2	0.29	0.17
ROUGE-L	0.23	0.14
BERTScore	0.24	0.27
FactCC	-0.06	-0.08
QuestEval	0.26	0.24
BARTScore	0.39	0.49
LongDocFACTScore	**0.61**	**0.61**

Table 3: Kendall's tau correlations between the human factual consistency annotations and automatic metrics for the LongSciVerify data set.

Metric	LongEval PubMed
ROUGE-1	0.15
ROUGE-2	0.26
ROUGE-L	0.22
BERTScore	0.18
FactCC	-0.14
QuestEval	0.13
BARTScore	0.22
LongDocFACTScore	**0.29**

Table 4: Kendall's tau correlations between the human factual consistency annotations and automatic metrics for the LongEval PubMed data set.

of metrics, the correlation is calculated between 45 summaries for each of the PubMed and ArXiv subsets of the LongSciVerify data set, and 40 summaries for the LongEval PubMed data set.

Table 3 gives Kendall's tau (Kendall, 1938) correlations[10] between the human measures of factuality and the automatic metrics for the LongSciVerify data sets. Table 4 gives the results of the same evaluation on the LongEval data set. Kendall's tau correlations were calculated, rather than Spearman correlations, due to being more robust for data sets with smaller sample sizes. A pairwise correlation matrix between the automatic metrics and human annotations of factuality is given in Figure 4. We restrict this plot to only the LongSciVerify PubMed data set, due to this being the data set which achieves the highest IAA score in Table 2, and is therefore likely to be the most reliable data set for evaluation. In Table 3, Table 4, and Figure 4, LongDocFACTScore, implemented by extending BARTScore, can be seen to correlate better with the human judgement of factual consistency than any other metric. Comparatively, we find that both FactCC and QuestEval show a low correlation with human judgement. BARTScore has a reasonable correlation with the human factual consistency annotations, however, since it is required to truncate the source document, we expect that it would become decreasingly correlated with human judgement as it is used to score texts of increasing length. ROUGE-2 and BERTScore perform best out of the baseline metrics evaluated, but no baseline metrics show a strong correlation with human mea-

[10] https://scipy.org